

RESEARCH/WORK EXPERIENCE

UNIVERSITY OF CAMBRIDGE

October 2026 - Present

Research Collaborator - Machine Learning

Remote

Cambridge, United Kingdom

- Conducting research with Prof. Mateja Jamnik and Xiangjian Jiang on robust and scalable prototype-based local feature selection for tabular data, extending the ProtoGate framework.
- Developing a scalable extension of ProtoGate that compresses its full exemplar bank into compact, class-aware representatives, progressing toward jointly learned, exemplar-anchored prototypes and local feature gates for interpretable prediction robust to outliers and label noise.
- Building a publication-oriented evaluation across synthetic and high-dimensional biomedical datasets, including METABRIC, benchmarking predictive fidelity, compression, feature-selection stability, computational efficiency, and contamination robustness toward an open-source release and peer-reviewed manuscript.

MILA - QUEBEC AI INSTITUTE

September 2026 - Present

Research Collaborator - Machine Learning

Montreal, Canada

- Conducting ongoing research with Prof. Guy Wolf and Adrien Aumon on categorical-aware sequential MCR forests for interpretable, unsupervised compression and reconstruction of mixed-type tabular data
- Building a reproducible Hydra/OpenML framework to characterize rate-distortion trade-offs across five forest methods, seven datasets, and multiple seeds; expanding the study through sequential-conditioning and categorical ablations toward a native mixed-data objective, an open-source benchmark, and a publication-ready research manuscript

INRIA, SODA TEAM

September 2025 - Present

Machine Learning Research Assistant

(full-time through January 2026, part-time/remote thereafter)

Palaiseau, France

- **First author "Disentangling Scale, Diversity, and Source in Pan-Cancer Survival Prediction Using TCGA and SEER", in preparation for submission to GigaScience, with Gaël Varoquaux and Judith Abécassis (INRIA SODA)**
- Quantifying the efficacy of pan-cancer vs. single-cancer training by harmonizing the TCGA (11k patients) and SEER (7M patients) cohorts, creating one of the largest unified clinical datasets for survival prediction
- Utilized advanced tabular representation learning (SurvivalBoost, TARTE, sentence embeddings) to evaluate model transferability across 31 shared clinical features
- Developed a robust data pipeline to resolve cross-cohort discrepancies, including sentinel normalization, TNM derivation, and patient-level deduplication

BAR-ILAN UNIVERSITY

June - September 2026

Machine Learning Research Assistant

Ramat Gan, Israel

- Conducted research with Prof. Ofir Lindenbaum on semi-supervised learning and foundation-model adaptation for tabular data under limited-label regimes.
- Explored and developed methods for extracting useful signal from unlabeled tabular data, including graph-based propagation, iterative pseudo-labelling, representation learning, and geometry-aware regularisation.
- Investigated semi-supervised adaptation strategies for tabular foundation models such as TabPFN and TabICL, incorporating self-training, Laplacian objectives, unlabeled-sample attention, and embedding-similarity constraints.
<https://github.com/dansuissa/tabular-foundation-ssl>

EPFL (École Polytechnique Fédérale de Lausanne)

June 2024 - September 2025

Machine Learning Research Assistant

(full-time through September 2024, part-time/remote thereafter)

Lausanne, Switzerland

- Worked under the supervision of Professor Pascal Fua, Romuald Houdré and Nicolas Villa in the Laboratory of Quantum and Nano-Optics (LQNO)
- Pioneered the application of time-series and tabular ML classifiers (XGBoost, MultiRocket, HIVE-COTE 2.0) for antimicrobial susceptibility testing, representing the first-ever use of these architectures with photonic crystal optical transmission data
- Engineered a custom pipeline to translate raw optical signals into high-accuracy biological sensing predictions : https://github.com/dansuissa/EPFL_AMR

INSERM

January - March 2025

Machine Learning Research Assistant (Bachelor Thesis)

Paris, France

- Worked under the supervision of Professor Guillaume Fillion, Ariel Lindner and Amir Pandi
- Extended a codon language model, CodonTransformer (nature.com/articles/), to optimize regulatory DNA sequences using novel tokenization and embedding techniques, enhancing biological sequence encoding : <https://github.com/dansuissa/CodonTransformerV2>

ÉCOLE POLYTECHNIQUE

September - December 2024

Undergraduate Machine Learning Research Assistant

Palaiseau, France

- In collaboration with Pr. Davide Buscaldi, evaluated LLM unlearning methods on the SemEval-2025 dataset to reduce memorization of sensitive data beyond output-filtering baselines

EDUCATION

MCGILL UNIVERSITY
MEng Artificial Intelligence

January 2026 - May 2027
Montreal, Canada

ECOLE POLYTECHNIQUE
BSc Mathematics and Computer Science

August 2022 - September 2025
Palaiseau, France

STANFORD UNIVERSITY
International Honors Program

June - August 2023
Stanford CA, USA

PROJECTS

- **Research Project - McGill University (8-month, part-time), supervised by Prof. Di Wu:** Benchmarking the trustworthiness of small (<10B-parameter) large language models across diverse hallucination scenarios, characterizing reliability and failure modes, and investigating training-free mitigation strategies for safer deployment.
- **Theory of Machine Learning Project:** Combined proof-based machine learning theory with empirical analysis to derive and validate scaling laws for overparameterized linear regression and two-layer neural networks, with a focus on SGD dynamics, generalization, and NTK, mean-field, and random-features limits

SKILLS

Python (TensorFlow, scikit-learn, skrub, PyTorch, numpy, pandas), R, C/C++, Git, Linux, Coq

LANGUAGES

- French (Native)
- English (Fluent)
- Spanish (Intermediate)
- Hebrew (Intermediate)

VOLUNTEERING & EXTRACURRICULARS

Scout movement group leader · Maths Olympiads tutor · IANTD Rescue Diver · MMA assistant coach